

Economics: Foundations and Models

ECON 101 - Macroeconomics

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- **1.1 Three Key Economic Ideas**
- 1.2 The Economic Problems All Societies Must Solve
- 1.3 Economic Models
- 1.4 Microeconomics and Macroeconomics
- 1.5 The Language of Economics

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What Is Economics? (1 of 2)

- People make **choices** as they try to attain their goals.
- Choices are necessary because we live in a world of **scarcity**.
- **Scarcity**: a situation in which unlimited wants exceed the limited resources available to fulfill those wants.
- **Economics** is the study of the choices people make to attain their goals, given their scarce resources.

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 - **Economic model**: a simplified version of reality used to analyze real-world economic situations.
- Typical economics questions:
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 - How do government spending and taxation affect the economy?
 - Why do governments control the prices of some goods?
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Everyday Examples of Scarcity

- **Your time:** cannot work, study, and relax at the same time.
- **Government budgets:** more spending on health care may mean less on education or infrastructure.
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- People respond to **incentives**.
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1. People Are Rational

- Economists generally assume that people are **rational**.
- **Rational**: people make choices based on what they believe will make them happy.
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Example: A \$5 Cup of Coffee

- You will not buy a \$5 cup of coffee unless you believe the benefit is worth at least:
 - the time and effort it takes you to earn \$5, **plus**
 - the value of any other item you could have bought with the \$5.
- **Hubbard-style interpretation:** Think in terms of **opportunity cost**:
 - If the best alternative use of \$5 is a bus pass or saving for tuition, you compare happiness from coffee vs. that alternative.

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2. People Respond to Incentives

- As **costs** and **benefits** change, so do people's actions.
- Example: Do Canadian banks choose to get robbed?
 - Bullet-resistant shields cost \$20,000.
 - Typical loss due to an armed robbery is \$1,200.
 - Armed robbery is rare.
 - **Result:** Many banks choose not to install bullet-resistant shields.

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More Incentive Examples

- **Students:** Higher course weight or harder marking may increase study time.
- **Firms:** A carbon tax raises the cost of emissions, encouraging cleaner technologies.
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3. Optimal Decisions Are Made at the Margin

- Most decisions involve doing a **little more** or a **little less** of something.
- Example: Should you watch an extra hour of TV or study instead?
- **Marginal cost (MC)**: the additional cost from a small increase in activity.
- **Marginal benefit (MB)**: the additional benefit from that small increase.
- Comparing MC and MB is called **marginal analysis**.

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Worked Example: Study vs. TV

Suppose:

- Extra hour of TV: $MB = 8$ “happiness points”, $MC = 0$ points.
- Extra hour of study: increases exam grade by 2 percentage points, which you value at 12 points of happiness (less stress, better GPA).
- **Decision:** If you switch from TV to study:
 - $MB(\text{study}) = 12$, $MC(\text{giving up TV}) = 8$.
 - Net gain = 4 points $\Rightarrow$ better to study the extra hour.

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Three Fundamental Questions

1.2 Learning Objective

Discuss how an economy answers these questions:

- **What** goods and services will be produced?
- How will the goods and services be produced?
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1. What Goods and Services Will Be Produced?

- All societies must decide which goods and services to produce.
- Producing more of one thing means producing less of another.
- This is a **trade-off**, resulting from the scarcity of productive resources.
- **Opportunity cost**: the value of the best alternative given up.
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2. How Will the Goods Be Produced?

- A firm may have multiple methods for producing the same output.
- Example 1 (Music production):
 - Hire a great singer and use standard production techniques; or
 - Hire a mediocre singer and use Auto-Tune to correct inaccuracies.
- Example 2 (Manufacturing labour):
 - As labour costs rise, firms may substitute capital (machines) for labour; or
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- **Market economy:** decisions of households and firms interacting in markets allocate resources.
- **Mixed economy:** most decisions are market-based, but government plays a significant role.
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- Markets may not result in fully efficient outcomes:
 - It takes time to discover the most efficient production methods.
 - Governments may interfere with market outcomes (taxes, controls).
 - Market outcomes can ignore those not directly involved:
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- **Positive analysis:** based on facts and logic.
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- Consider a government **cap** on house prices.
- Potential benefits:
 - Lower home prices for buyers.
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 - Fewer houses built.
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